NTNU – Trondheim
Norwegian University of
Science and Technology

TET 4565 -Electricity Markets and Energy System Planning, Specialisation Course

Hydropower Scheduling Hierarchy and Water Values

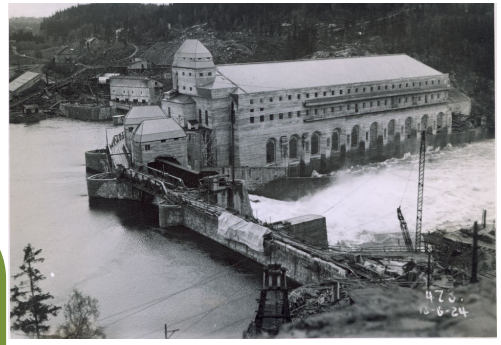


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Professor
Autumn 2026

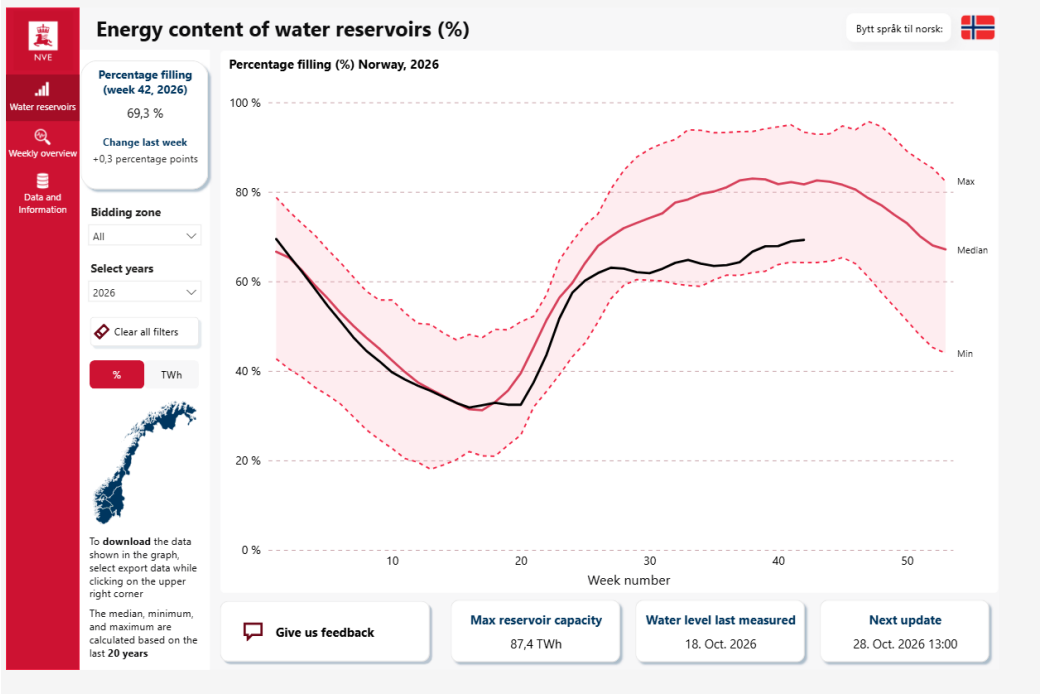
Overview

Hydropower Scheduling Hierarchy

Water Value Calculation







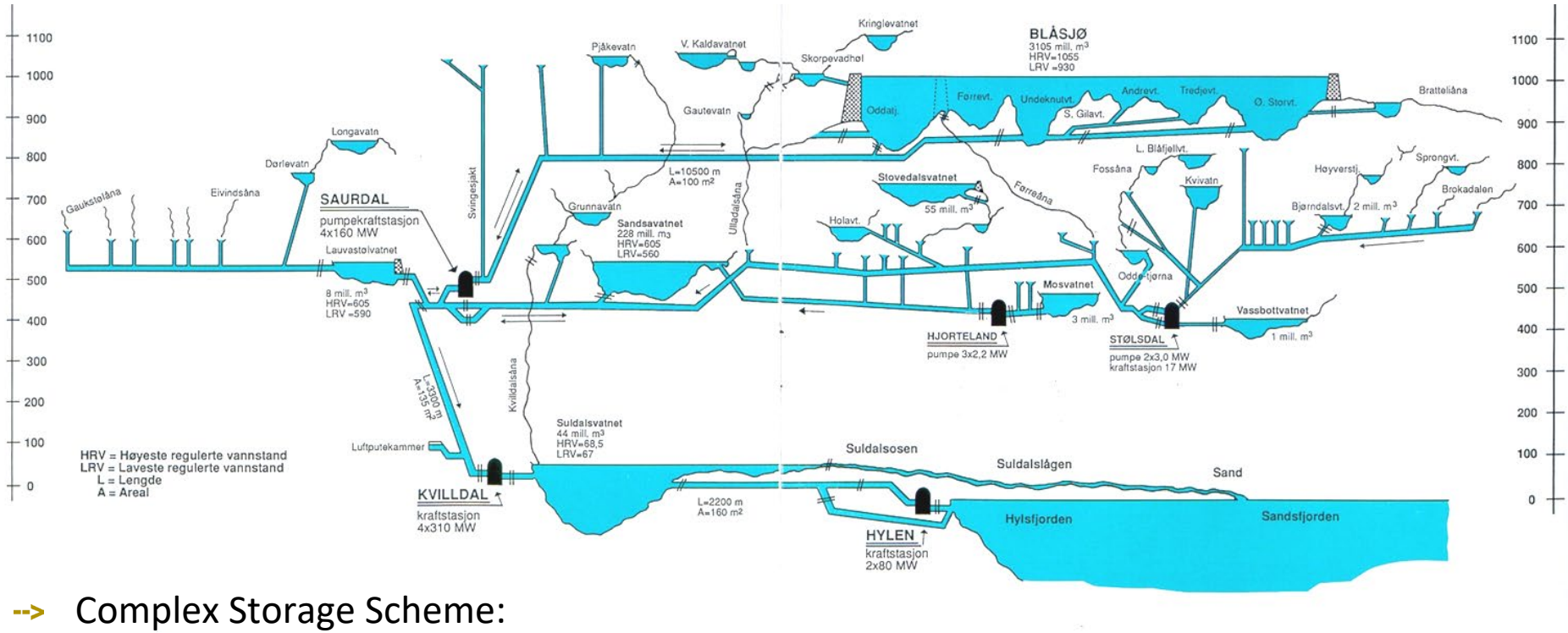
The figures represent 97.5 percent of the total reservoir capacity. The total reservoir capacity is 87.4 TWh. h. Min, max and median values for the period 2000 - 2019.
Source: [The Norwegian Water Resources and Energy Directorate (NVE)]

References

- Compendium in hydro power scheduling
- Scheduling Toolchains in Hydro-Dominated Systems
- Found in the “Learning materials” folder on the Canvas.



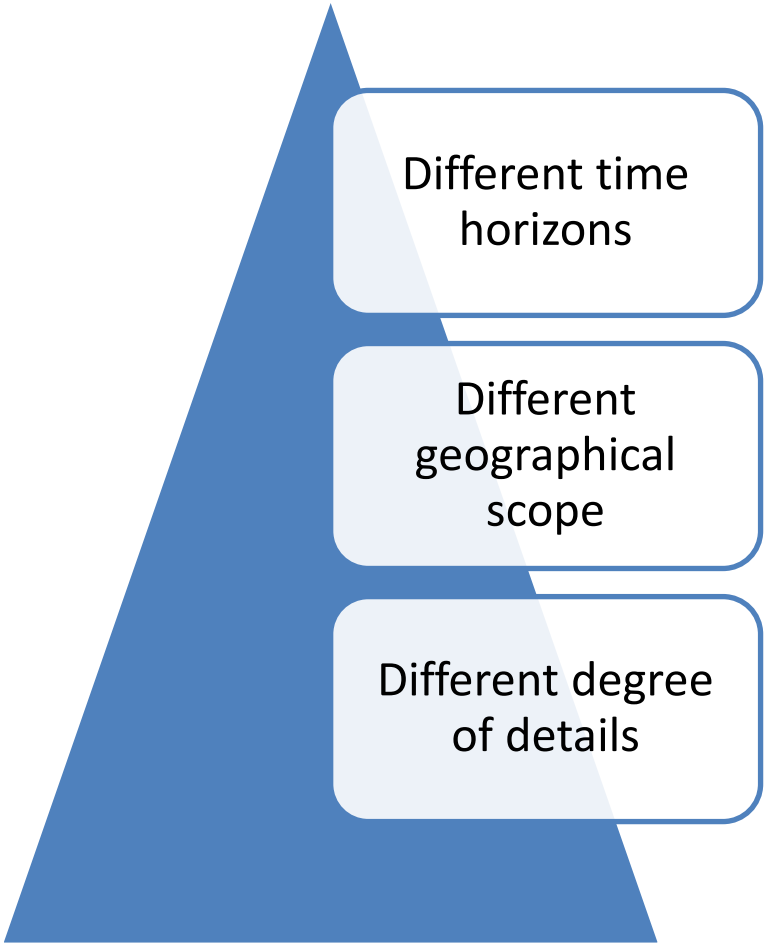
Storage and waterways-Real life example

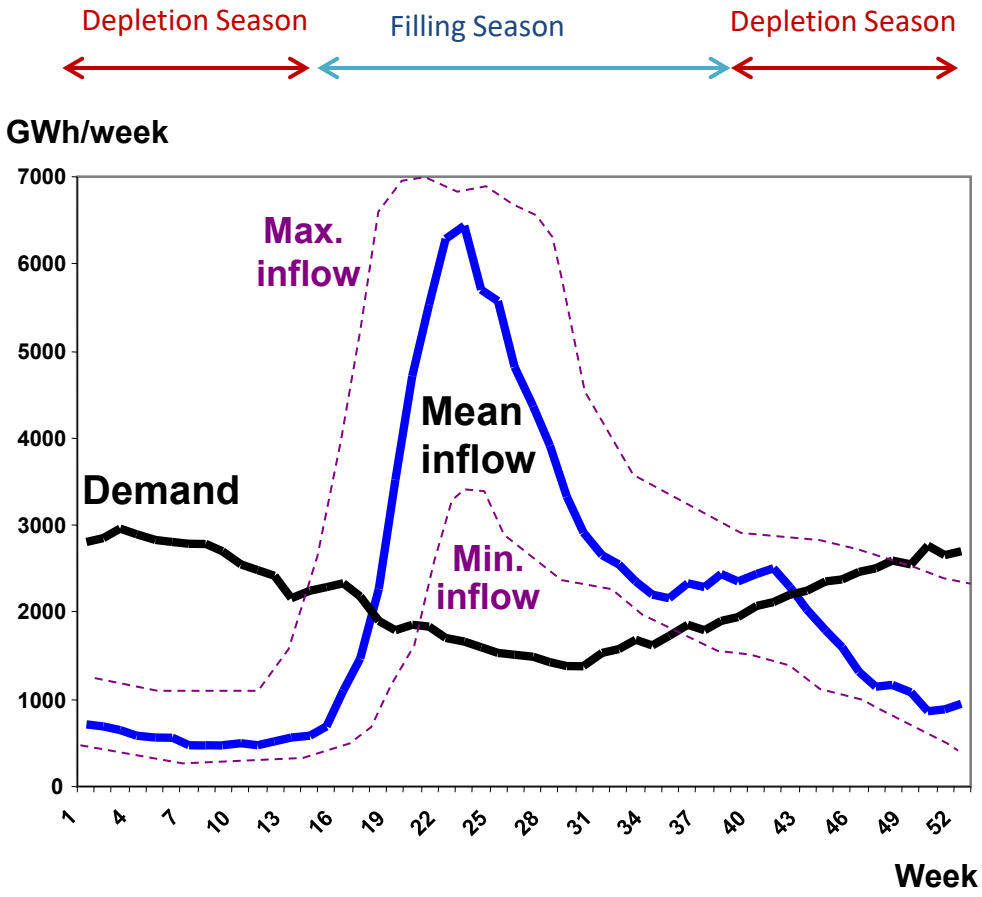


--> Complex Storage Scheme:

- 1 Major reservoir, contains water for multi-year production (in case of dry year(s))
- 24 smaller reservoirs that are channeled into the system
- 3 Major Power plants (all underground), and 2 pumping stations

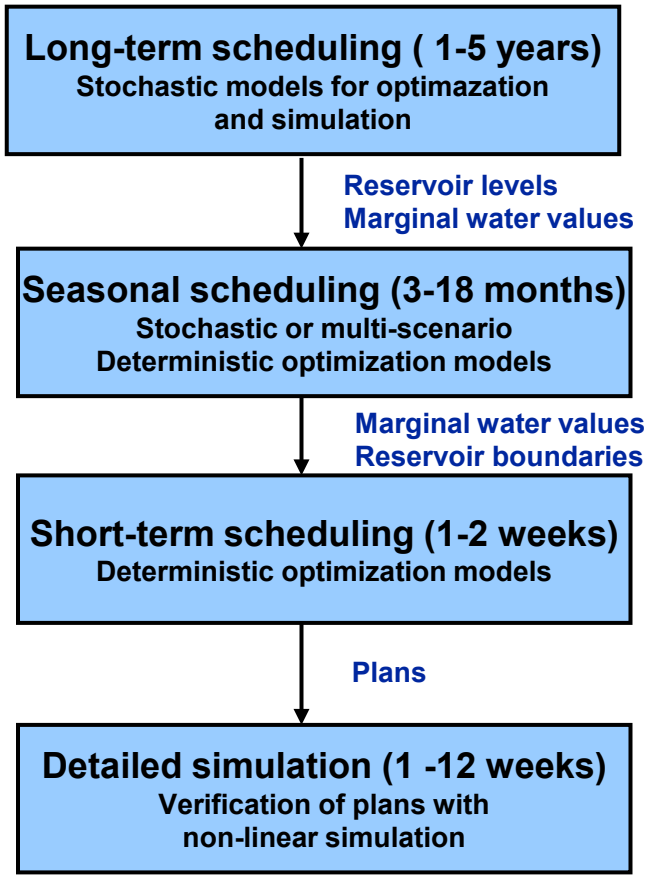
Scheduling Task



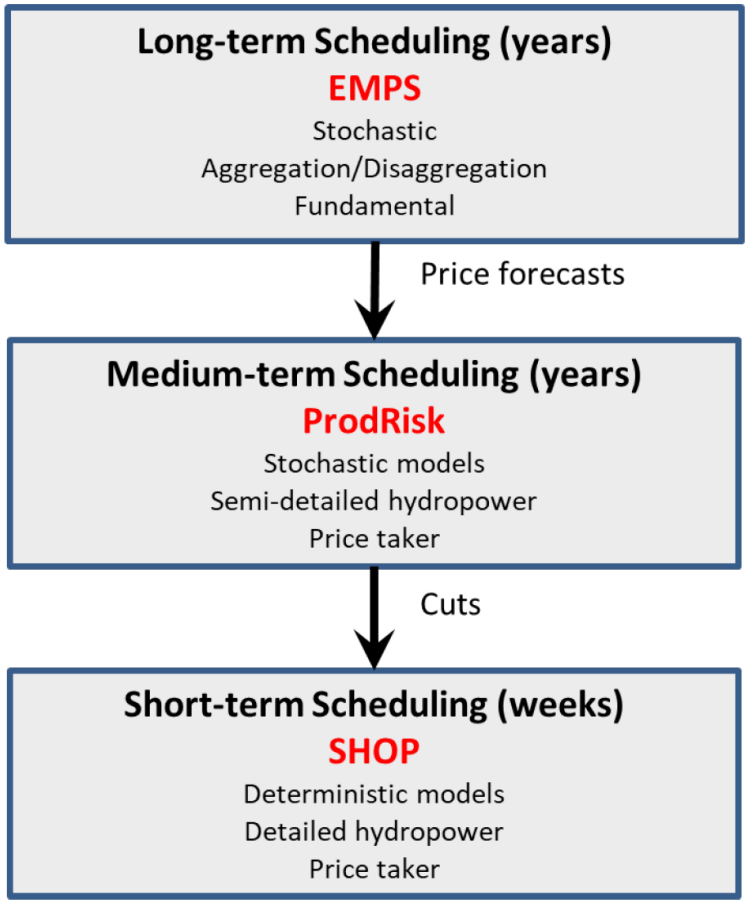


Scheduling Hierarchy

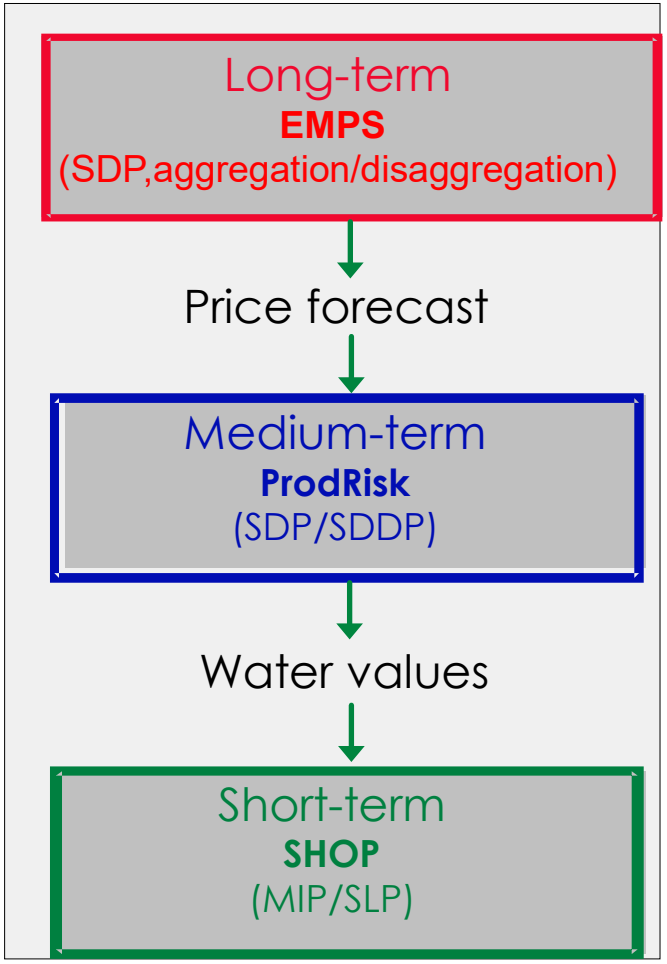
Compendium



A. Helseth and A. C. Geber de Melo, "Scheduling Toolchains in HydroDominated Systems: Evolution, Current Status and Future Challenges for Norway and Brazil," SINTEF Energy Research, Tech. Rep., 2020, https://sintef.brage.unit.no/sintef-xmlui/handle/11250/2672581?locale-attribute=en .



Scheduling Hierarchy



Applications long-term:
- System analyses
- Expansion planning
- Maintenance planning

Applications short-term:
- Portfolio bidding
- Production plans

- Shorter horizon
- Less uncertainty
- System boundary decreases
- More details

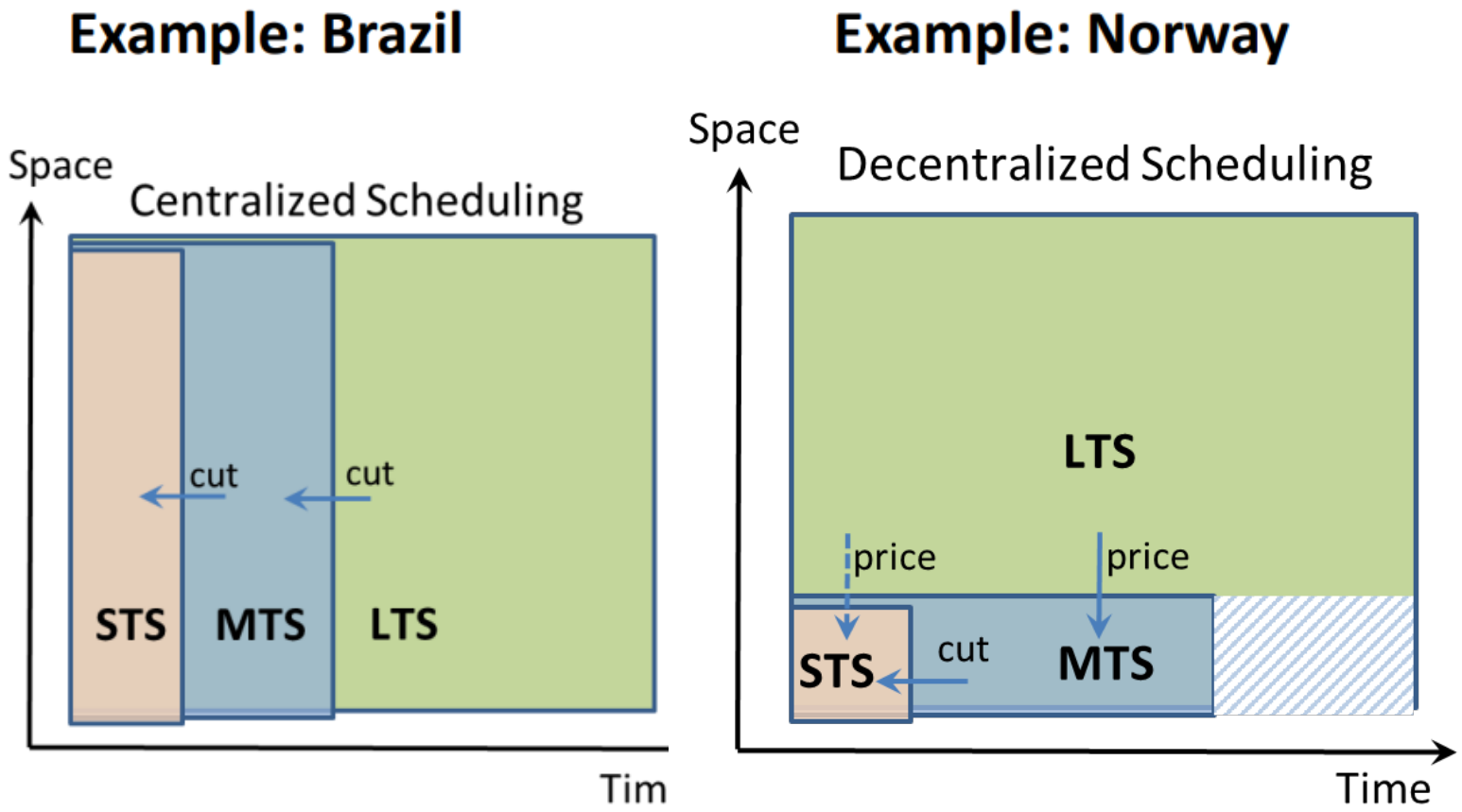
A. Helseth and A. C. Geber de Melo, "Scheduling Toolchains in HydroDominated Systems: Evolution, Current Status and Future Challenges for Norway and Brazil," SINTEF Energy Research, Tech. Rep., 2020, https://sintef.brage.unit.no/sintef-xmlui/handle/11250/2672581?locale-attribute=en .

Two (equivalent) objectives

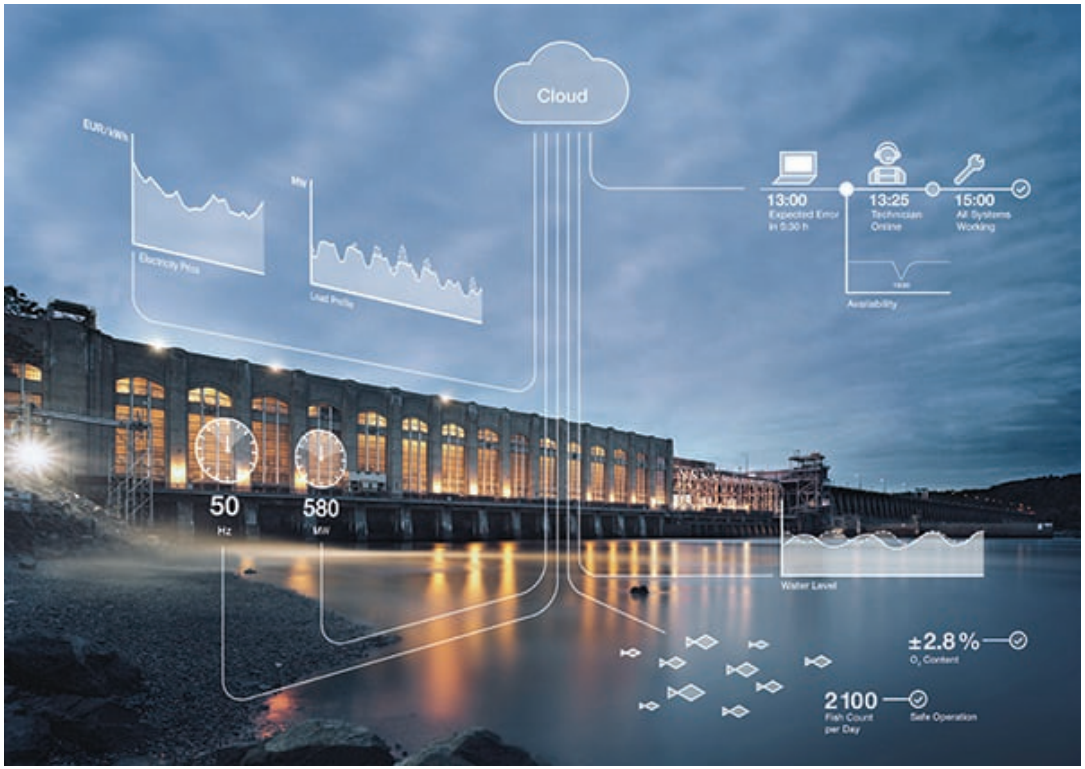
- Cost minimization
  - Relevant in centralized markets (Brazil, Iceland) and Norway before deregulation (1991)
  - Objective: minimize expected operational costs over time

- Profit maximization
  - Relevant in decentralised markets
  - Market simulation: Prices
  - Profit maximization per company

Geographical scopes



Water Values

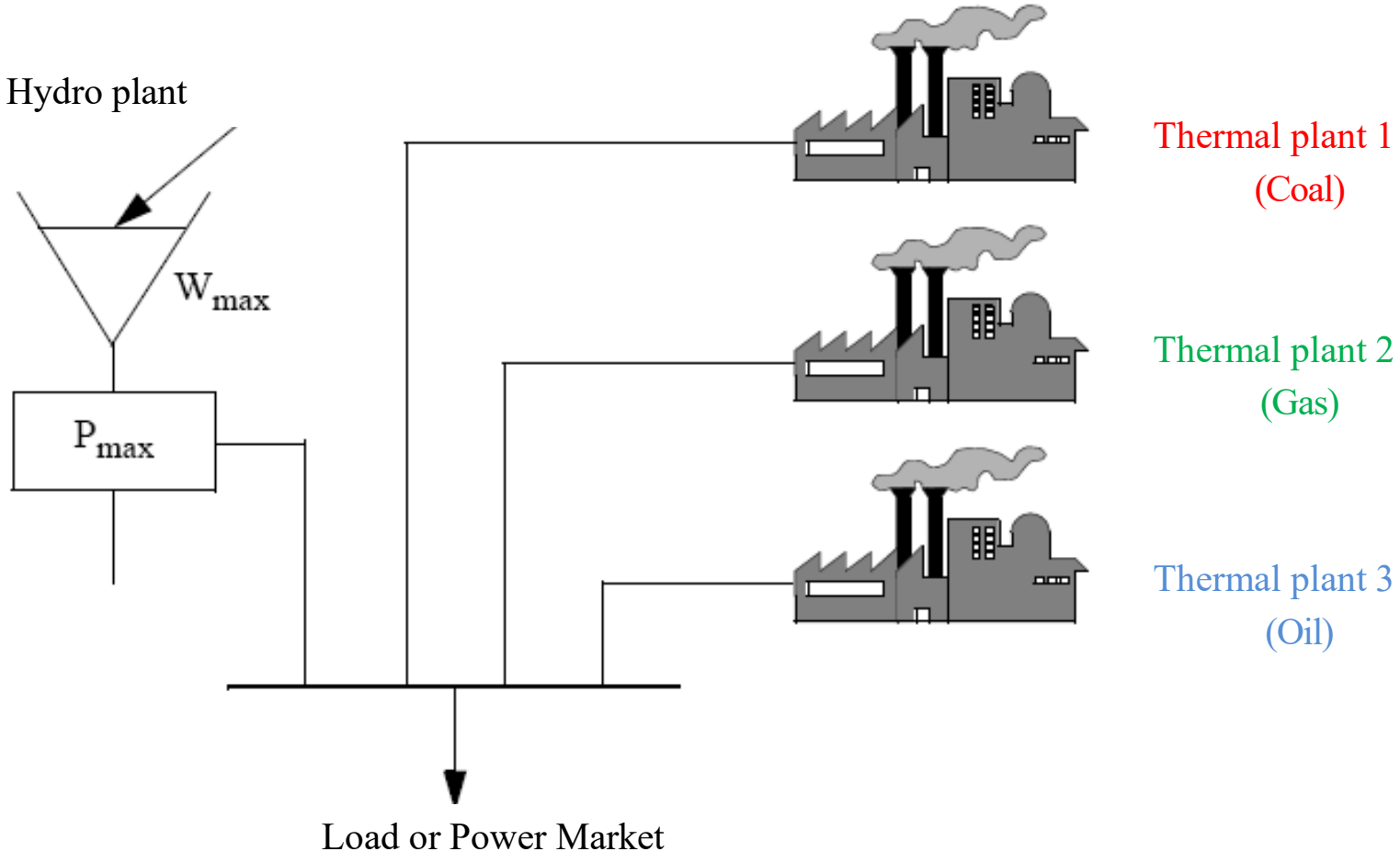


Water value – Definition

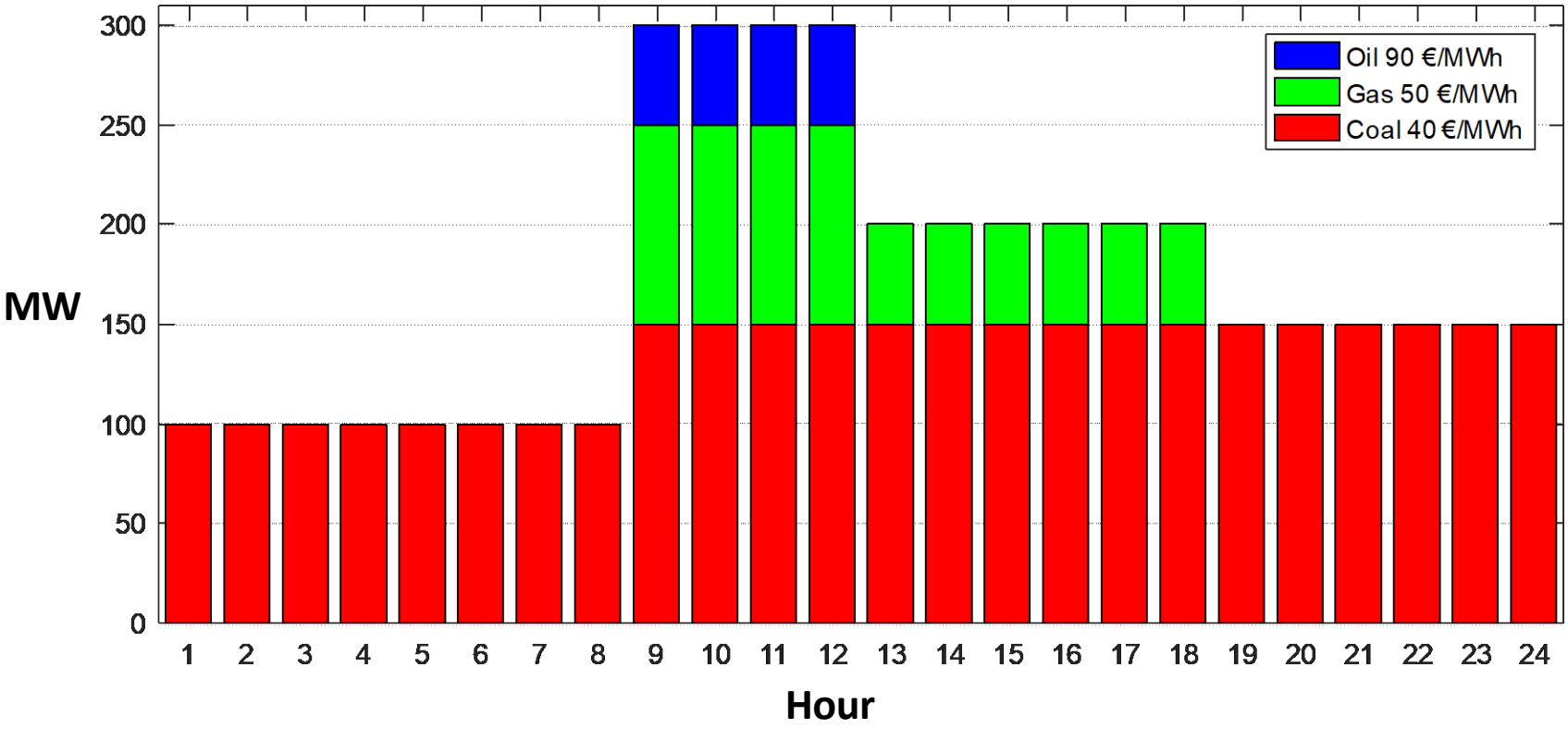
The ***water value*** is the ***expected marginal value*** of having an additional unit of water in the reservoir

We will use Stochastic Dynamic Programming to find the water value

Deterministic Example hydro-thermal system

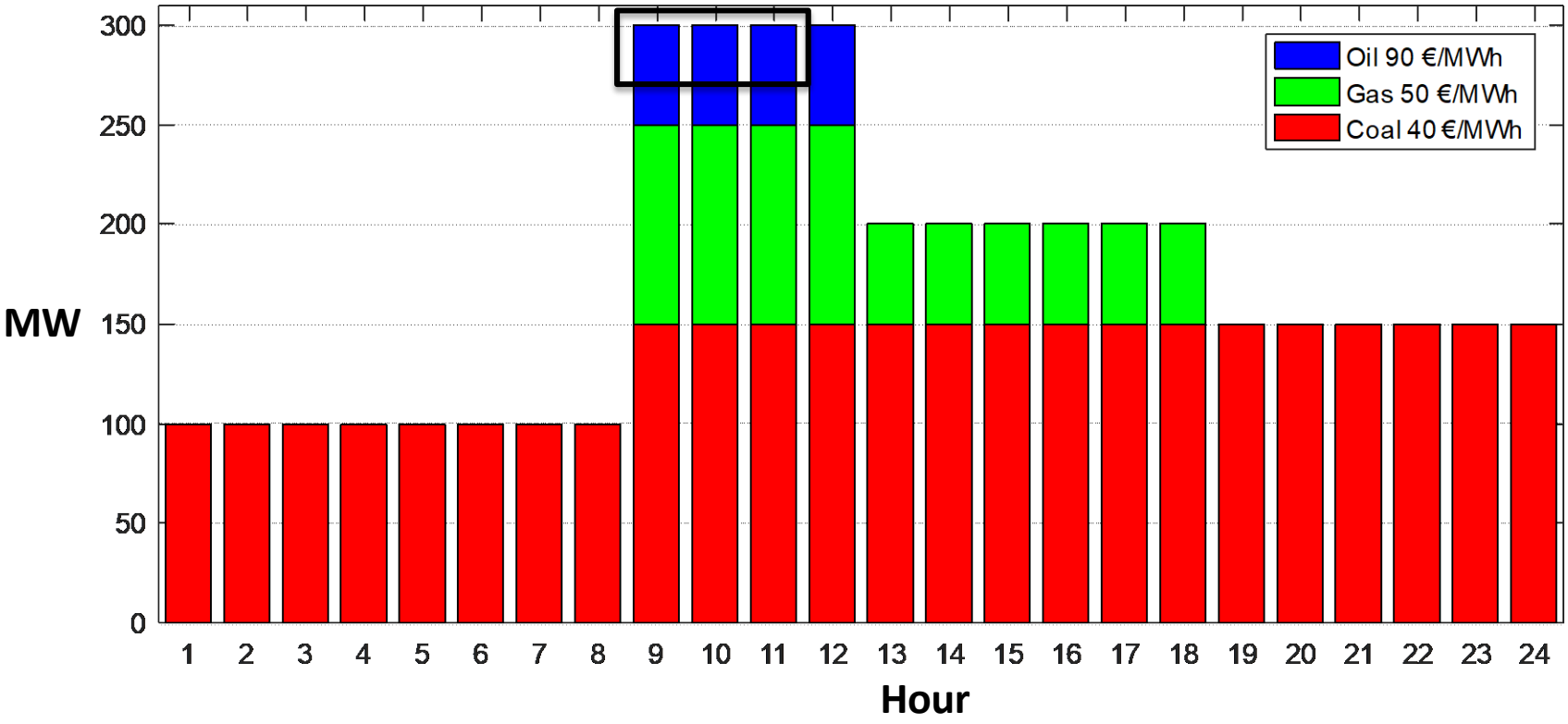


Water Value Calculation-Example



- Hydro: 25 MW, 75 MWh
  - How to use it optimally?
  - Marginal value of having 1 more MWh?

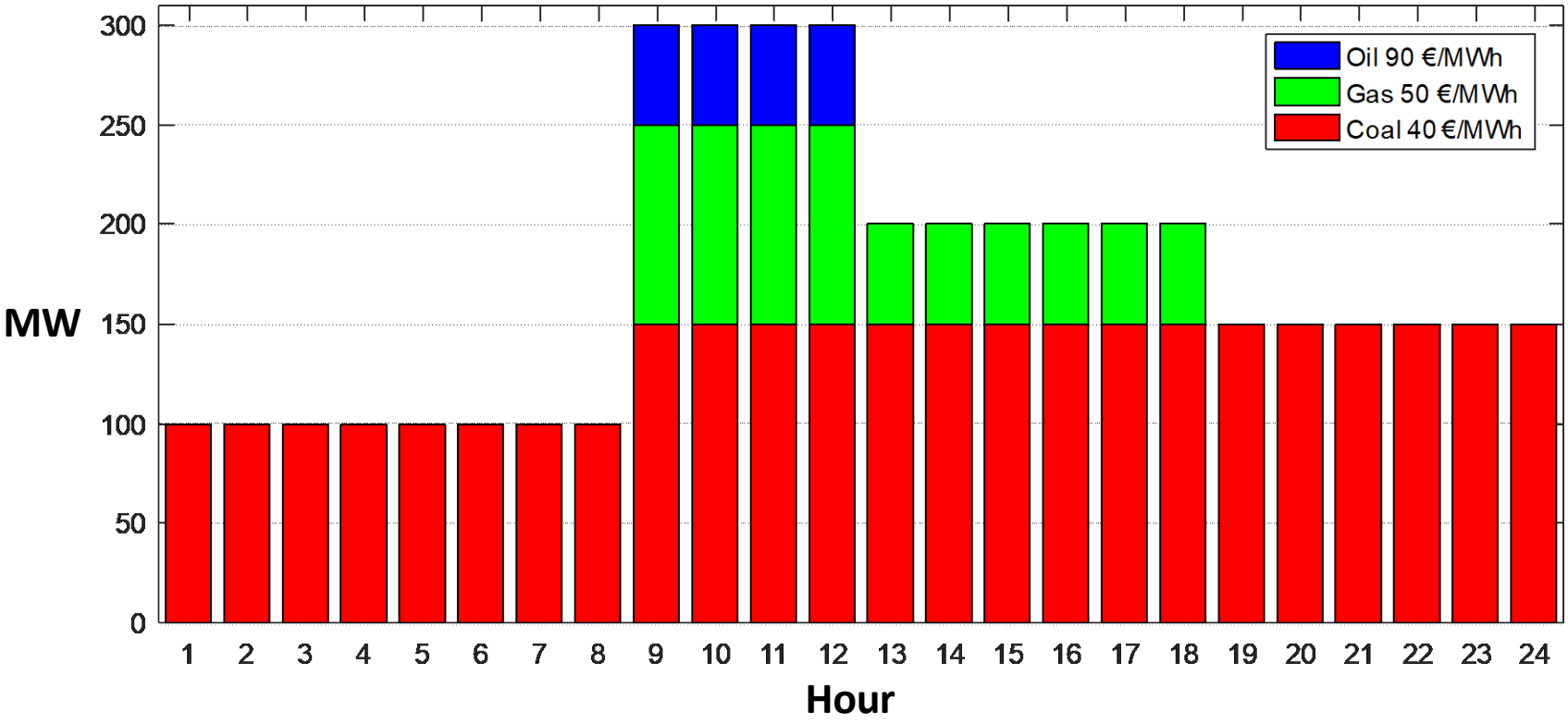
Water Value Calculation-Example (1)



- Hydro: 25 MW, 75 MWh
  - How to use it optimally?
  - Marginal value of having 1 more MWh?

Hydro replaces Oil -> Water value is 90 €/MWh

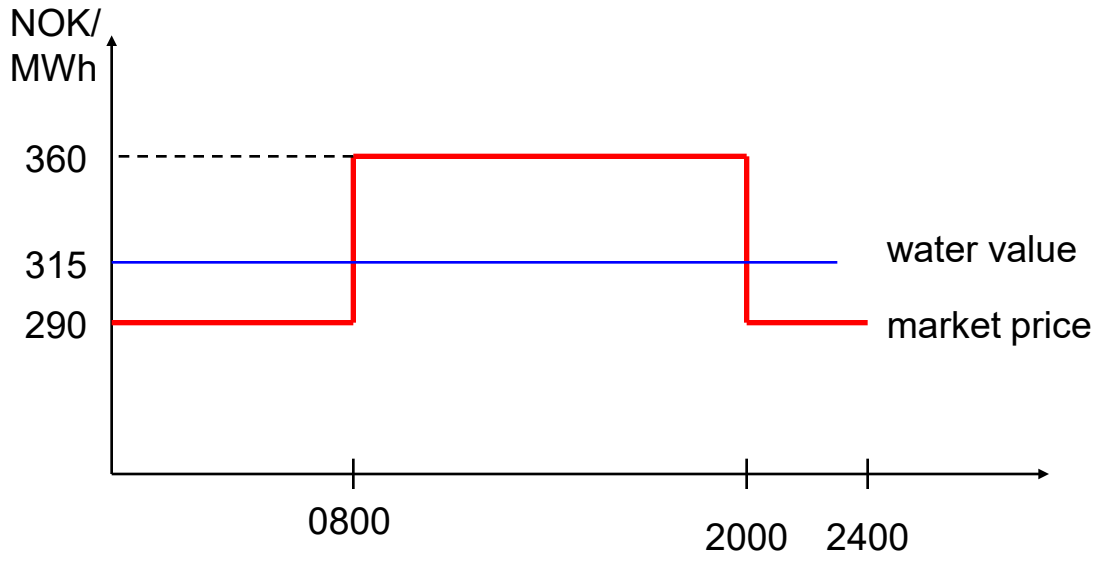
Water Value Calculation-Example (2)



- Find the water value of the following hydro power specifications:
  - A) 25 MW power capacity and 100 MWh stored energy 50
  - B) 25 MW power capacity and 300 MWh stored energy 40
  - C) 100 MW power capacity and 150 MWh stored energy 90

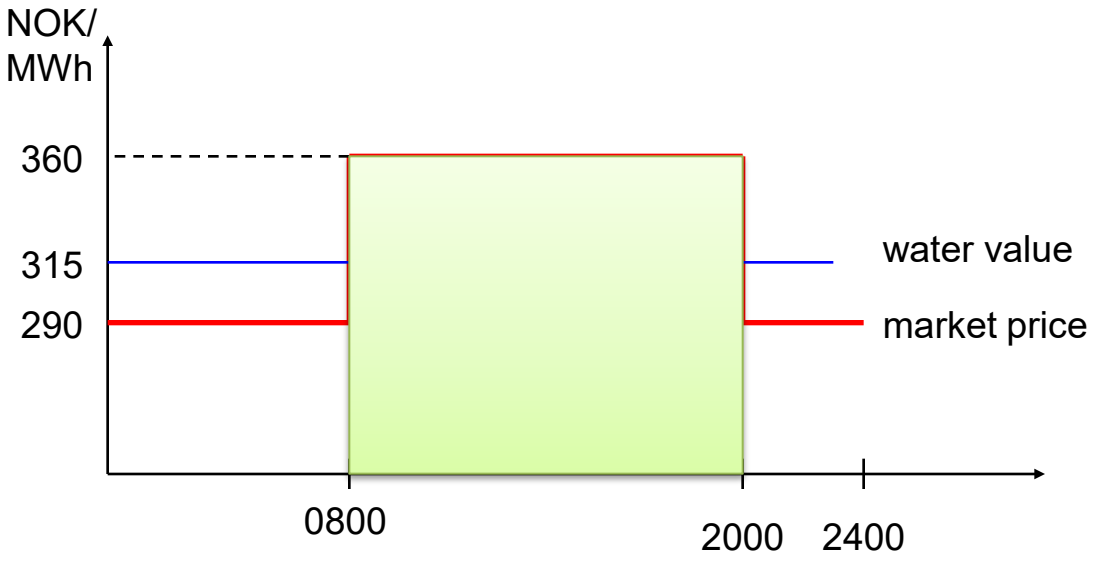
Optimal Hydropower Dispatch– example

- Water value: 315 NOK/MWh
- Plant capacity 100 MW
- Market price:

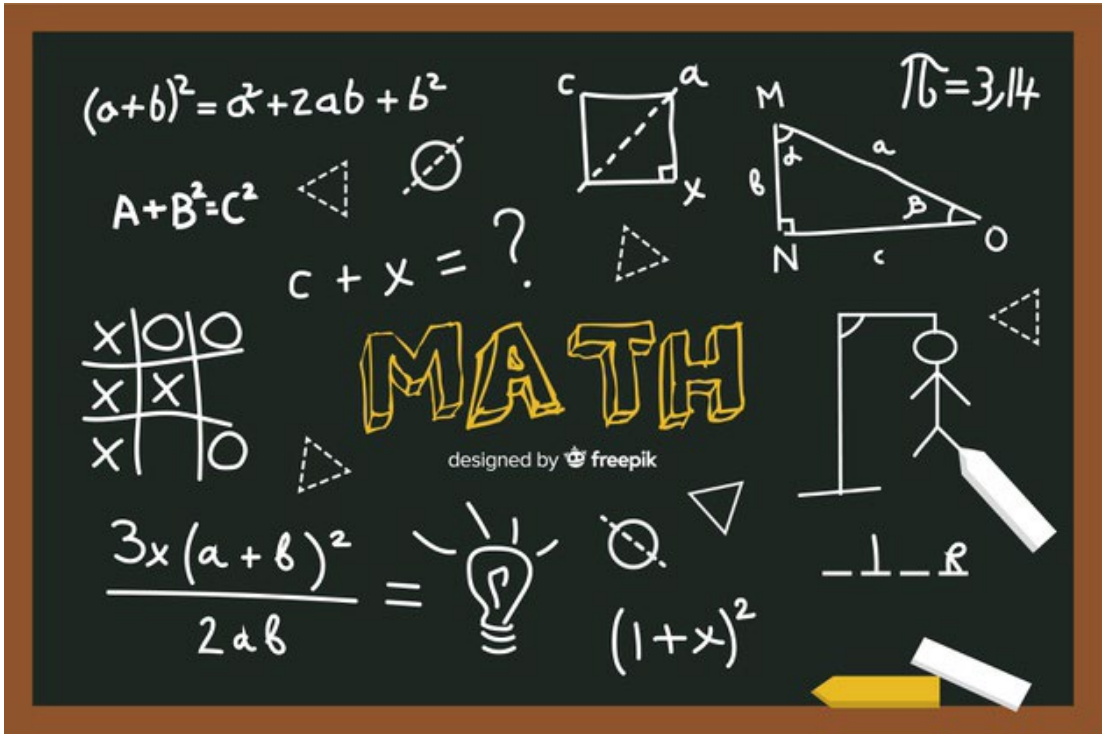


Optimal Hydropower Dispatch– example

- Water value: 315 NOK/MWh
- Plant capacity 100 MW
- Market price:



Water Value Calculation



Mathematical background-recap from previous lecture

Consider the following two-stage deterministic problem:

First-stage cost

$$\min_{x_1, x_2} \; c_1 x_1 + c_2 x_2$$

Second-stage cost

$$\text{subject to}$$

$$A_1 x_1 \geq b_1$$

First-stage constraint

$$E_1 x_1 + A_2 x_2 \geq b_2$$

Linking constraint

Mathematical background

Solve the first-stage and the second-stage problems separately

Step 1) **Discretize** x_1 into trial values

$\{ \hat{x}_1^i, i = 1, \dots, n\}$

Step 2) For each trial value, solve the second-stage problem→ obtain the optimal future cost

Step 3) Construct the future-cost function $\alpha_1(x_1)$→ use interpolation between trial points

Step 4) Solve the first-stage problem using $\alpha_1(x_1)$→ choose x_1 by balancing present and future cost

First-stage problem:

$$\min_{x_1} \; c_1 x_1 + \alpha_1(x_1)$$

subject to

$$A_1 x_1 \geq b_1$$

Second-stage problem:

$$\alpha_1(x_1) = \min_{x_2} \; c_2 x_2$$

subject to

$$A_2 x_2 \geq b_2 - E_1 x_1$$

Dynamic programming (DP)

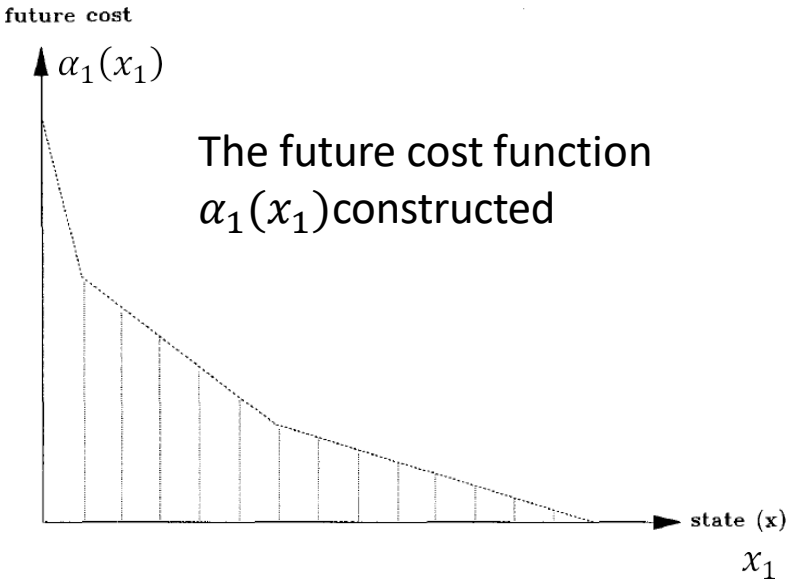
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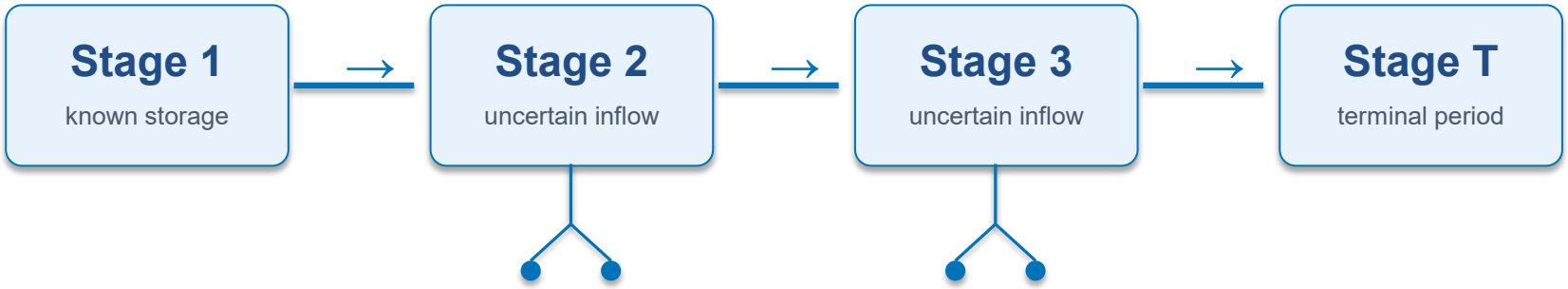
Step 4) Solve the first-stage problem using $\alpha_1(x_1)$→ choose x_1 by balancing present and future cost



The future cost function $\alpha_1(x_1)$constructed

From two-stage DP to multi-stage SDP

Repeat the same future-cost calculation for every week and reservoir level



Different inflow outcomes → different next-period storage levels

← **BACKWARD RECURSION: calculate future costs from T to 1**

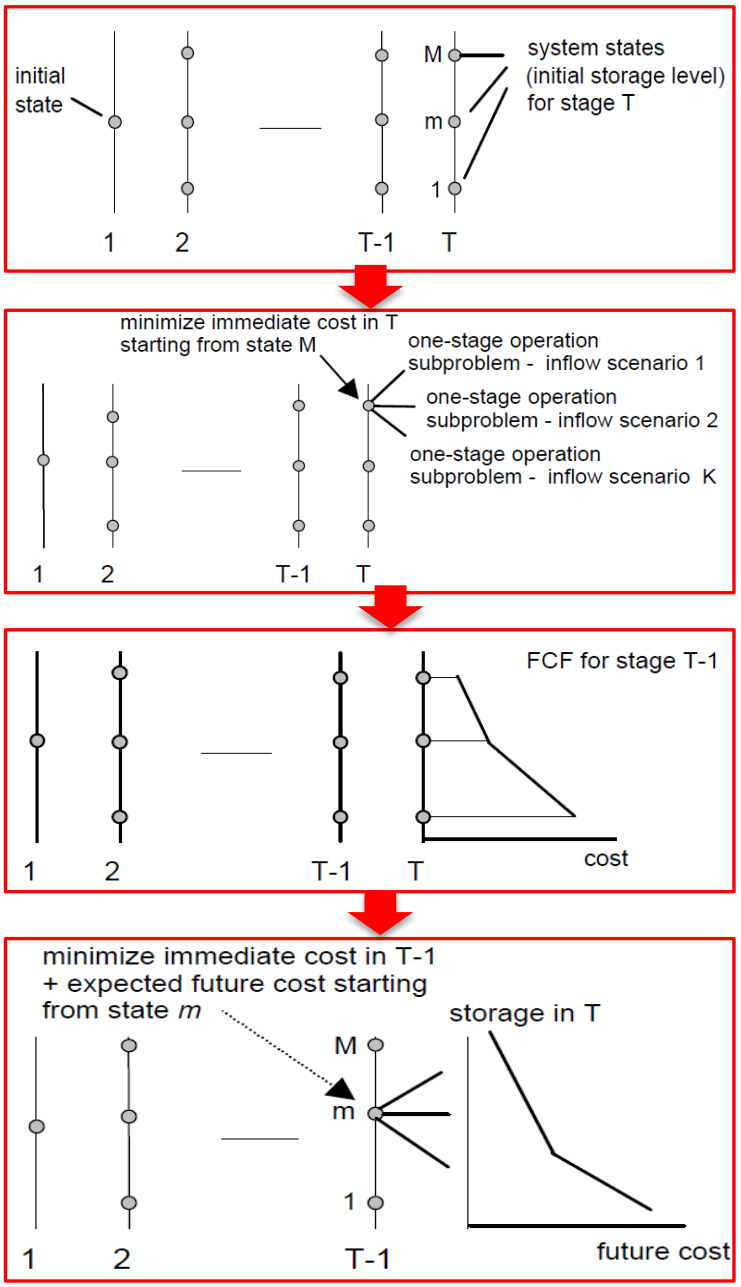
1 Discretize states	2 Solve last stage	3 Move backward	4 Check the convergence
Choose storage levels for each stage.	For each storage level and inflow, minimize operating cost.	Current cost + expected future cost; construct FCF.	The converged value function gives the long-run operating policy.

At each stage: $F_t^{(k)}(v_t) = \min_{u_t} \left\{ c_t(v_t, u_t) + \mathbb{E}\left[F_{t+1}^{(k)}(v_{t+1}) \right] \right\}$

SDP Recursion

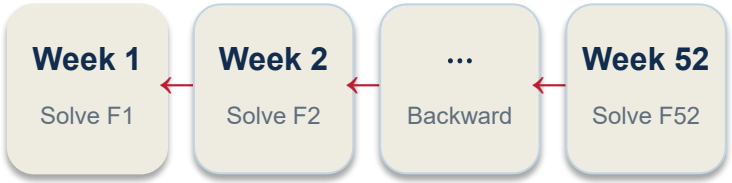
- **Step a)** Definition of system states for each stage *t*, discretize the system states (e.g., reservoir storage levels) into several segments to represent different operating conditions. The initial state is assumed to be known.
- **Step b)** Optimal Strategy Calculation - Last Stage
- **Step c)** Calculation of the FCF for Stage T-1
- **Step d)** Backward recursion for previous stages: Calculation of Operation Costs for Stage T-1 and FCF for Stage T-2
- **Step e)** Continue the backward recursion until the values of the Future Cost Function (FCF) in the first and last stages (or consecutive iterations) are approximately equal.

After convergence: apply the resulting policy in a separate forward simulation.



SDP: one backward sweep vs. annual iterations

WITHIN ONE ANNUAL SWEEP



At every week: evaluate all discretized storage states and inflow outcomes.

BETWEEN ANNUAL SWEEPS

Iteration 0 Assume terminal FCF

↓

Iteration 1 Backward weeks 52 → 1

↓

Iteration 2 Update terminal FCF; repeat

Repeat until marginal water values stabilize

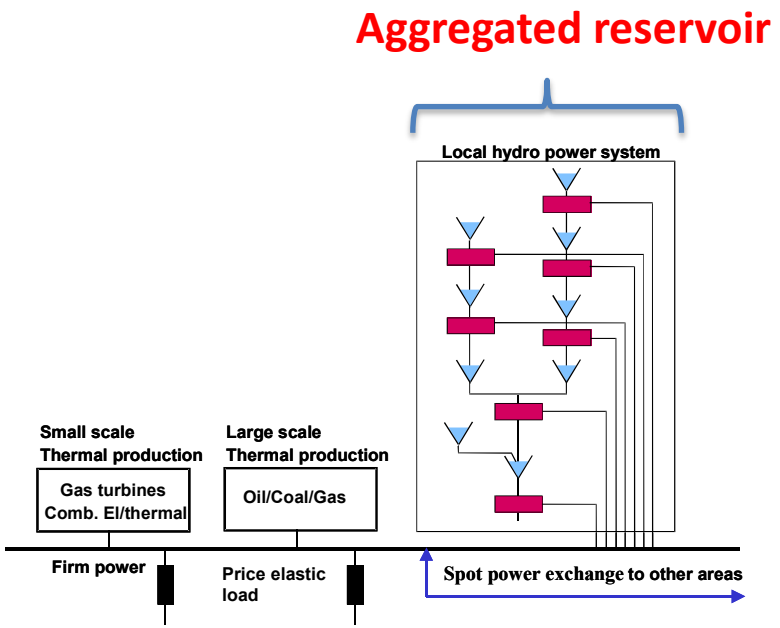
Cyclic update $F_{\text{term}}^{(k+1)}(v) \leftarrow F_1^{(k)}(v)$

Stopping test: $\max_v \left| F_{\text{term}}^{(k+1)}(v) - F_{\text{term}}^{(k)}(v) \right| < \varepsilon$ then the forward simulation follows convergence.

Water value method

Stochastic Dynamic Programming (SDP)

- **Stage:** time – one week
- **State:** reservoir volume (discretized, 51 levels)
- **Return function:** operation dependent cost
- **Decision:** hydro production (energy "withdrawal" from reservoir)



Nomenclature

Sets

t – timesteps [weeks]

Parameters

C^g- Cost of grid import

C^r- Cost of rationing

D_t - Consumption

Stochastic variables

ι_t - inflow [MWh/h]

Decision variables

g_t- grid import [MWh/h]

r_t- rationing [MWh/h]

p_t- hydro production [MWh/h]

x_t - reservoir level at start of period [Mwh]

s_t- spill [Mwh]

Functions

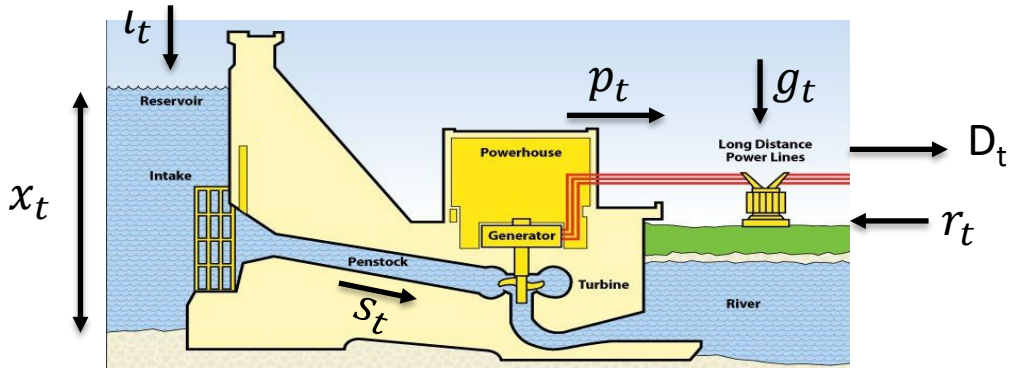
FC – future cost

Dual multipliers

λ_t - marginal cost of operation

μ_t – marginal value of stored water

Problem formulation for deterministic single stage



$$\min_{g,r,x,p,s\geq 0} J_t = C^g g_t + C^r r_t + FC_{t+1}(x_{t+1})$$ Objective function

Subject to:

$$p_t + g_t + r_t = D_t \qquad (\lambda_t)$$ Power balance constraint

$$X_t - p_t - s_t + \iota_t = x_{t+1} \qquad (\mu_t)$$ Reservoir balance constraint

$$g_t \leq G^{max}$$

$$p_t \leq P^{max}$$

$$X^{min} \leq x_t \leq X^{max}$$

SDP for Hydropower

Goal:

Optimize multi-stage reservoir operation under uncertain inflows (ι_t), and demands (D_t).

Stage *t*: Decision problem

$$\min_{g,r,x,p,s\geq 0} J_t = C^g g_t + C^r r_t + FC_{t+1}(x_{t+1})$$

Subject to:

$$p_t + g_t + r_t = D_t \qquad (\lambda_t)$$
$$X_t - p_t - s_t + \iota_t = x_{t+1} \qquad (\mu_t)$$
$$g_t \leq G^{max}, \quad p_t \leq P^{max}, \quad X^{min} \leq x_t \leq X^{max}$$

Algorithm (pseudocode):

1) Initialization: Set terminal value $V_{T+1}(x_{T+1}) = FC_{T+1}(x_{T+1})$
2) Backward recursion: For $t = T \rightarrow 1$, for all reservoir levels x_t compute:

$$V_t(x_t) = \min_{g,r,x,p,s\geq 0} \left[C^g g_t + C^r r_t + \mathbb{E}_{\iota_t} [V_{t+1}(x_{t+1})] \right]$$

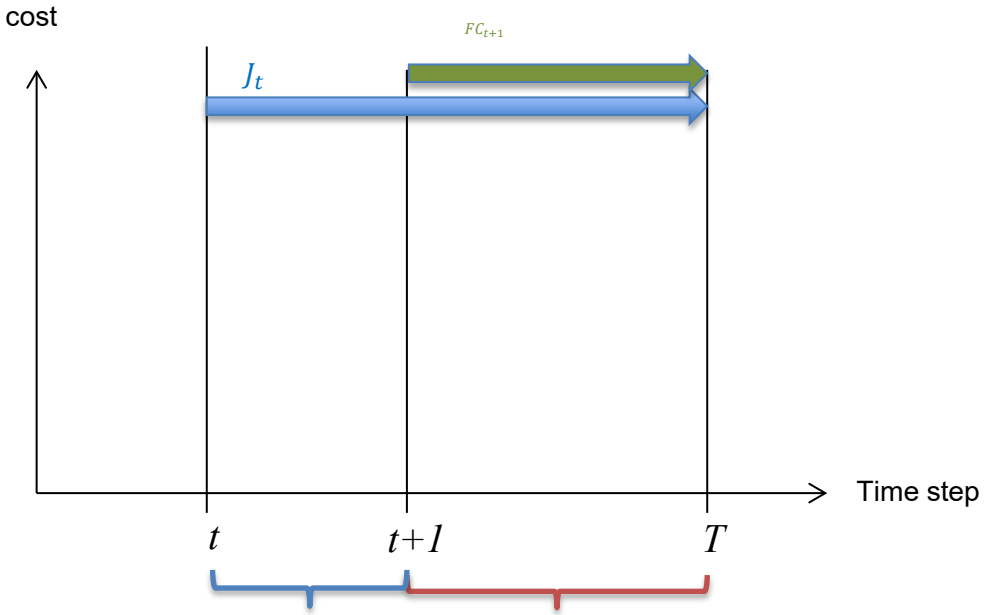
3) Cyclic update: $V_{T+1}^{k+1}(x_{T+1}) = V_1^k(x_1)$ and Stopping test where we have stable water values:

$\max_v \left| V_{T+1}^{(k+1)}(v) - V_{T+1}^{(k)}(v) \right| < \varepsilon$,where *k* is iteration number

4) Forward simulation: Given inflow realizations ι_t , apply optimal policy to evaluate energy production, revenue, and final storage.

Outputs: Optimal releases p_t^*, r_t^*, s_t^* ,value functions $V_t(x_t)$, and marginal water values $\frac{\partial V_t}{\partial x_t}$.

Objective function



$$\min_{g,r,x,p,s\geq 0} J_t = C^g g_t + C^r r_t + FC_{t+1}(x_{t+1})$$

$$\min_{g,r,x,p,s\geq 0} J_t = C^g g_t + C^r r_t + J_{t+1}$$ Bellman formulation

Formulate the optimisation problem as a Lagrangian problem

$$\min_{g_t, r_t, p_t, x_{t+1}, \lambda_t, \mu_t} L = \begin{cases} C^g g_t + C^r r_t + FC_{t+1}(x_{t+1}) \\ +\lambda_t (p_t + g_t + r_t - D_t) \\ +\mu_t (X_t - p_t - s_t + \iota_t - x_{t+1}) \end{cases}$$

- The Karush-Kuhn-Tucker condition says that either the dual multiplier must be 0, or the restriction must be binding.
- The power balance and reservoir balance constraints are added with their associated multipliers.

Water value - calculation

- Water value = the marginal value of having one extra unit stored in the reservoir
- Equivalent to μ_t the dual multiplier of the reservoir balance constraint.
- Note that μ_t refers to the reservoir filling of time t.

Hydro power system optimality

$$\frac{\partial L}{\partial p_t} = \lambda_t - \mu_t = 0$$

$$\frac{\partial L}{\partial x_{t+1}} = \frac{\partial FC_{t+1}}{\partial x_{t+1}} - \mu_t = 0$$

$$\frac{\partial FC_{t+1}}{\partial x_{t+1}} = \lambda_t$$

- The water value has to be equal to the marginal cost of operation
- In a perfect competitive market equivalent to
  - Price = water value at end of period
- Trade off: Use water now for lower cost, or save water for lower cost later
- Note: Assuming no other binding constraints.

Envelope Theorem (sketch of proof)

The Envelope Theorem tells you that the partial derivative of the optimal value of the objective function with respect to a parameter is equal to the partial derivative of the objective function with respect to that parameter, evaluated at the optimal values of the decision variables.

$$F(x) = \min_{x} \{ f(x,u) \}$$
$$s.t. \quad g(x,u) = 0 \quad (\lambda)$$

Postulate $\Longrightarrow$ $\dfrac{\partial F}{\partial u} = \dfrac{\partial f^*}{\partial u} + \lambda \dfrac{\partial g^*}{\partial u}$ *and* $x = x(u)$

Sketch of proof:

$$L = f(x,u) + \lambda g(x,u) \qquad f.o.c.: \quad \frac{\partial L}{\partial x} = \frac{\partial f^*}{\partial x} + \lambda \frac{\partial g^*}{\partial x} = 0$$

$$\frac{\partial F}{\partial u} = \frac{\partial L}{\partial u} = \frac{\partial f}{\partial u} + \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial \lambda}{\partial u} g + \lambda \left(\frac{\partial g}{\partial u} + \frac{\partial g}{\partial x}\frac{\partial x}{\partial u} \right) =$$

$$\frac{\partial f}{\partial u} + \frac{\partial \lambda}{\partial u} \overset{=0}{g} + \frac{\partial x}{\partial u} \left(\overbrace{\frac{\partial f}{\partial x} + \lambda \frac{\partial g}{\partial x}}^{=0} \right) + \lambda \frac{\partial g}{\partial u} = \underline{\underline{\frac{\partial f}{\partial u} + \lambda \frac{\partial g}{\partial u}}}$$

Q.E.D.

(indirect derivatives cancel out)

Envelope Theorem

Using the sketch of proof of the envelope theorem, we get

$$\min_{g,r,x,p,s\geq 0} J_t = C^g g_t + C^r r_t + FC_{t+1}(x_{t+1})$$

s.t:

$$p_t + g_t + r_t = D_t \qquad (\lambda_t)$$
$$X_t - p_t - s_t + \iota_t = x_{t+1} \qquad (\mu_t)$$ (which is the only place X_t occurs)

Derivation and the envelope theorem than gives:

$$\frac{\partial J_t}{\partial x_t} = 0 + \mu_t \Rightarrow \frac{\partial J_t}{\partial x_t} = \mu_t$$

$$\frac{\partial J_t}{\partial x_t} = \mu_t \text{ and for } t+1\text{: } \frac{\partial J^*_{t+1}}{\partial x_{t+1}} = \mu_{t+1} \Rightarrow \frac{\partial FC^*_{t+1}}{\partial x_{t+1}} = \mu_{t+1}$$

System optimization – optimality for hydropower

Optimality condition

$$\frac{\partial FC_{t+1}}{\partial x_{t+1}} = \lambda_t$$

Water value next period = power price

Envelope theorem applied on t+1 optimization problem

$$\frac{\partial FC^*_{t+1}}{\partial x_{t+1}} = \mu_{t+1}$$

(indirect derivatives cancel out)

Optimality condition (next period)

$$\frac{\partial L}{\partial p_{t+1}} = \lambda_{t+1} - \mu_{t+1} = 0 \Rightarrow \lambda_{t+1} = \lambda_t$$

Same price next period

without other binding constraints!

Influence of reservoir restrictions I

$$\min_{g,r,x,p,s\geq 0} C^g g_t + C^r r_t + FC_{t+1}(x_{t+1})$$

Subject to

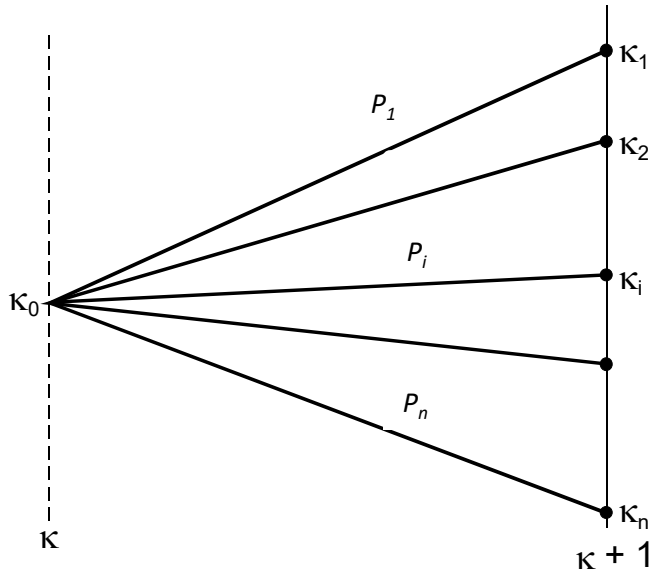
$$p_t + g_t + r_t = d_t$$
$$X_t - p_t - s_t + \iota_t = x_{t+1}$$
$$g_t \leq G^{max}$$
$$p_t \leq P^{max}$$
$$X^{min} \leq x_t \leq X^{max}$$

If these restrictions are binding, the lagrangian will have another non-zero term

Note that the reservoir constraint and production limitation drive a wedge between the prices in weeks *t* and *t+1*

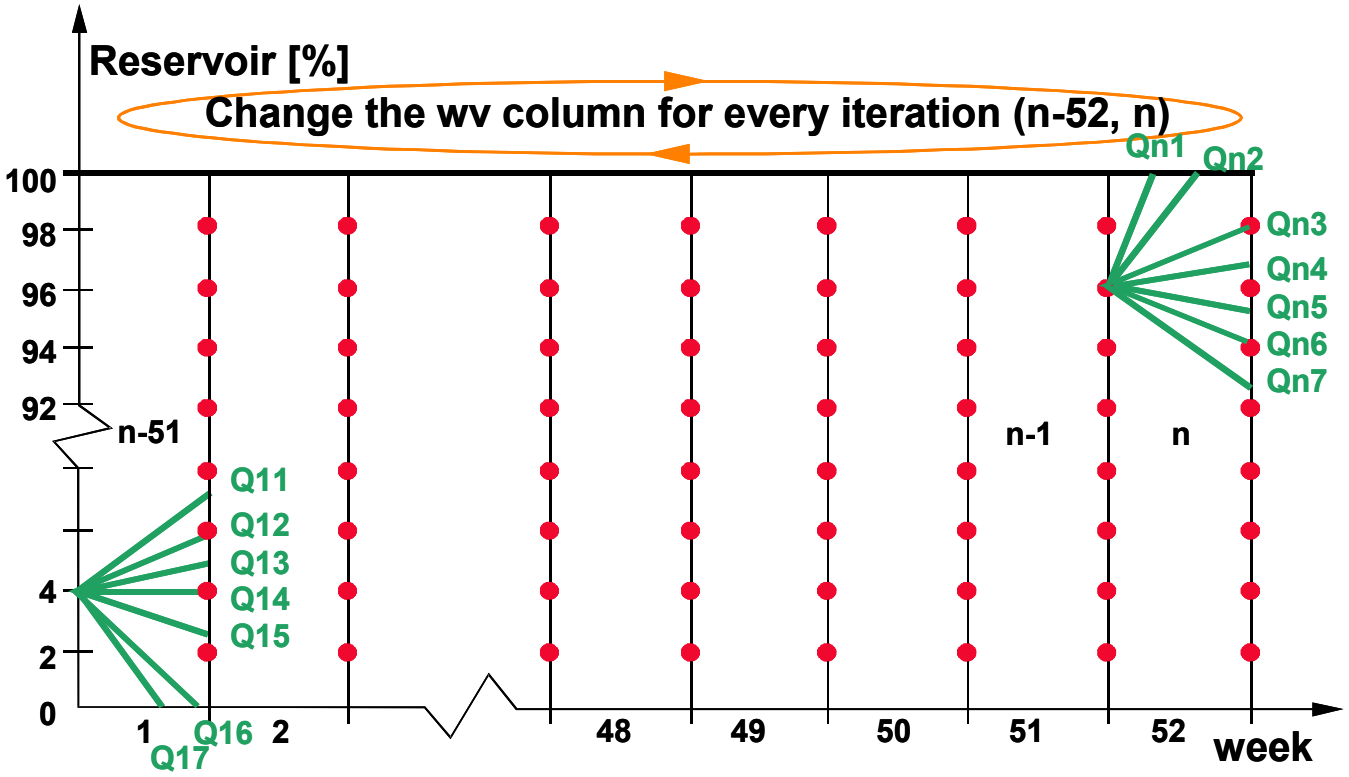
Stochastic inflow

$$k_0 = \sum_{i=1}^{n} p_i k_i$$

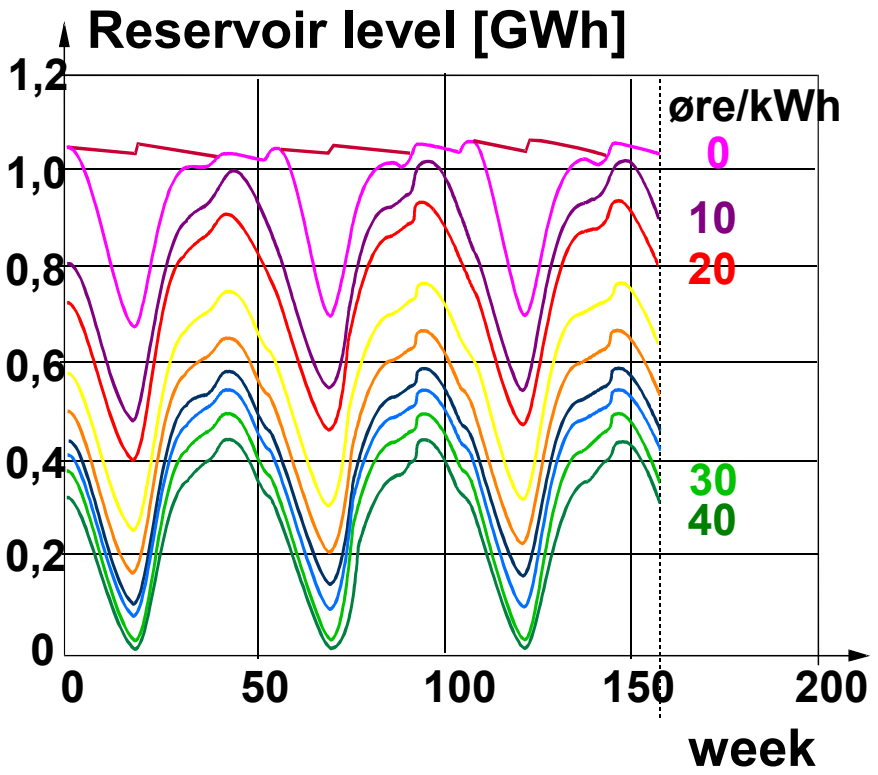


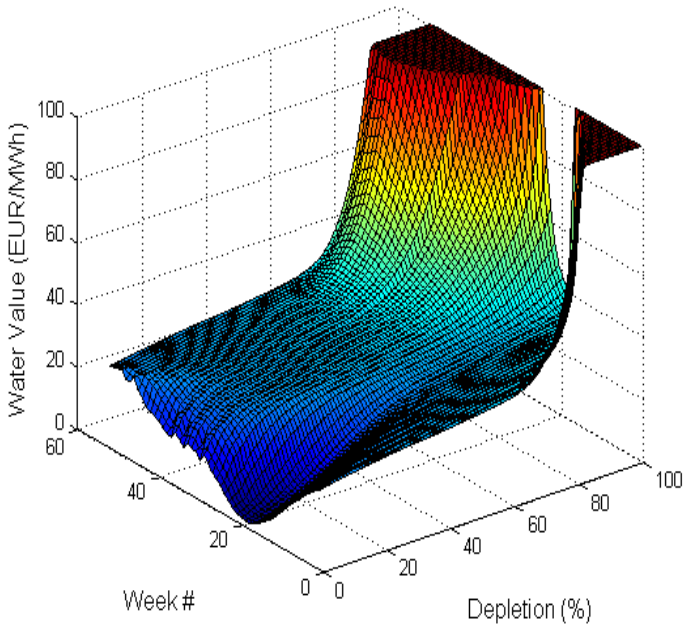
$$\sum_{i=1}^{n} p_i = 1{,}0$$

Water value calculations for discrete reservoir levels



Water value example
iso-price curves





Conclusions

- Hydropower scheduling operates within a hierarchical framework:
  - ✓ Long-term → strategic resource planning
  - ✓ Short-term → operational optimization and market participation
- The water value represents the marginal value of stored energy, the cornerstone linking economic and physical decision layers.
- Stochastic Dynamic Programming (SDP) provides a systematic way to derive water values through backward recursion over discretized reservoir states and inflow scenarios.
- Final water values provide a foundation for bidding strategies, system planning, and portfolio optimization in hydro-dominated markets